BSIT 375: Knowledge Discovery and Data Science

Week 1 Lecture Notes: Foundations and Descriptive Statistics

Based on Larose & Larose: *Discovering Knowledge in Data (2nd Ed)*

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1: Introduction

1.1 Knowledge Discovery (KDD) and Data Mining

To understand this field, we must distinguish between the broad goal of finding information and the specific tools used to get there.

1. Knowledge Discovery (KDD)
Knowledge Discovery in Databases is the **entire process** of turning raw, messy data into high-level, actionable insights. It is a multi-step journey that starts with a business question and ends with a decision.

Simple Example: Imagine a supermarket chain looking at millions of receipts. The **KDD process** is the whole project of cleaning that receipt data, analyzing it, and finally discovering that people who buy diapers on Friday nights also tend to buy beer. The "knowledge" gained is the strategy to place these items closer together to increase sales.

2. Data Mining
Data Mining is a **specific step** within the KDD process. It is the phase where we apply mathematical algorithms to the data to find patterns. If KDD is the process of mining for gold, Data Mining is the actual act of using the drill to find the vein of ore.

Simple Example: Using an algorithm to group customers into "high spenders" and "bargain hunters" based on their past 12 months of transactions is **Data Mining**.

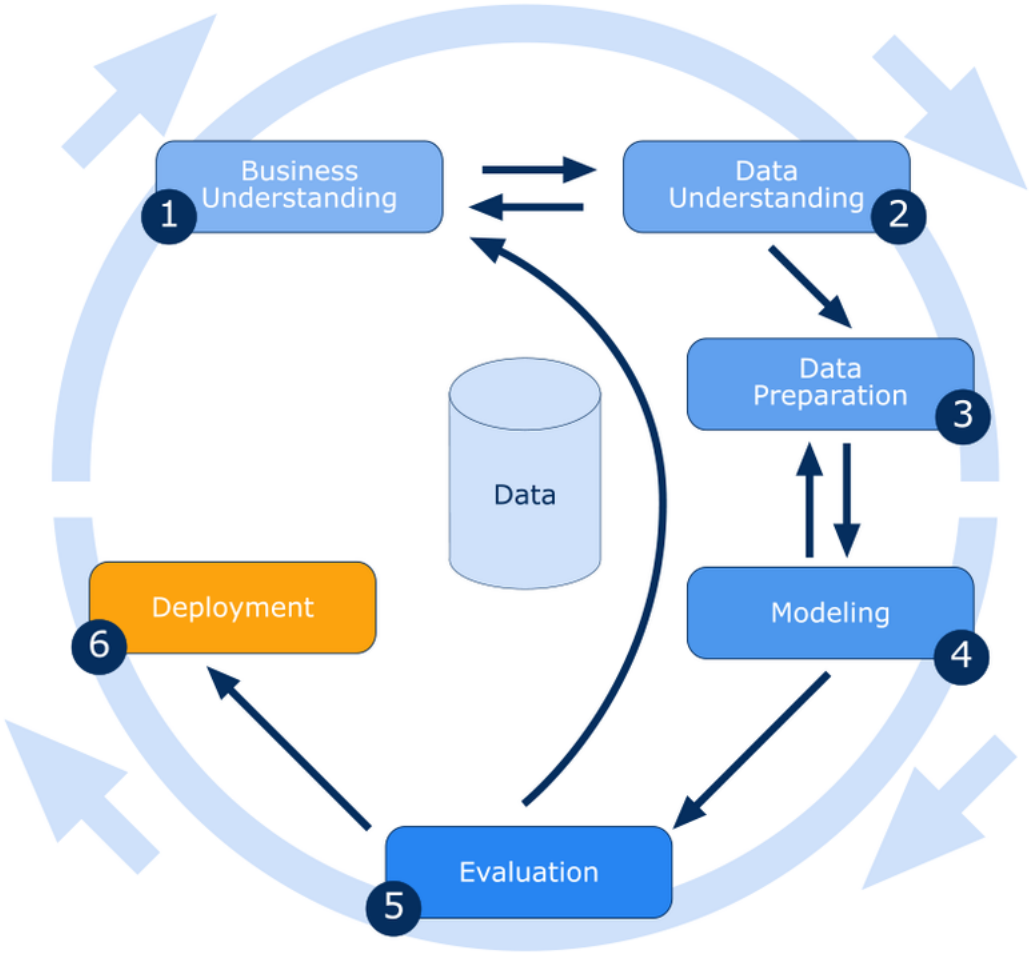
3. Data Science
While KDD is focused on finding patterns, **Data Science** is a broader discipline. It includes KDD but also involves the engineering required to build data pipelines, the software development to deploy models, and the infrastructure (like Cloud Computing) needed to store massive datasets.

- **Data:** "40% of our users are in Asia." (A raw fact).
- **Information/Pattern:** "Users in Asia are more active between 8 PM and 11 PM." (Discovered via Data Mining).
- **Knowledge:** "By scheduling our marketing notifications for 8 PM in Asian time zones, we can increase click-through rates by 25%." (The actionable result of KDD).

2: The KDD Landscape and Industry Methodologies

1.2 CRISP-DM: The Industry Standard

The **Cross-Industry Standard Process for Data Mining (CRISP-DM)** is the most widely adopted framework. It treats data mining as a cyclical process rather than a linear one.



1. **Business Understanding:** What problem are we solving? (e.g., reducing churn).
2. **Data Understanding:** Exploring the Churn or Adult datasets to see what we have.
3. **Data Preparation:** Cleaning and transforming (Handling missing values).
4. **Modeling:** Applying algorithms.
5. **Evaluation:** Does the model meet business goals?
6. **Deployment:** Putting the model into production.

1.3 SEMMA: The SAS Institute Alternative

A common alternative to CRISP-DM is **SEMMA**. While CRISP-DM focuses on the business context, SEMMA focuses more on the technical manipulation of the data.

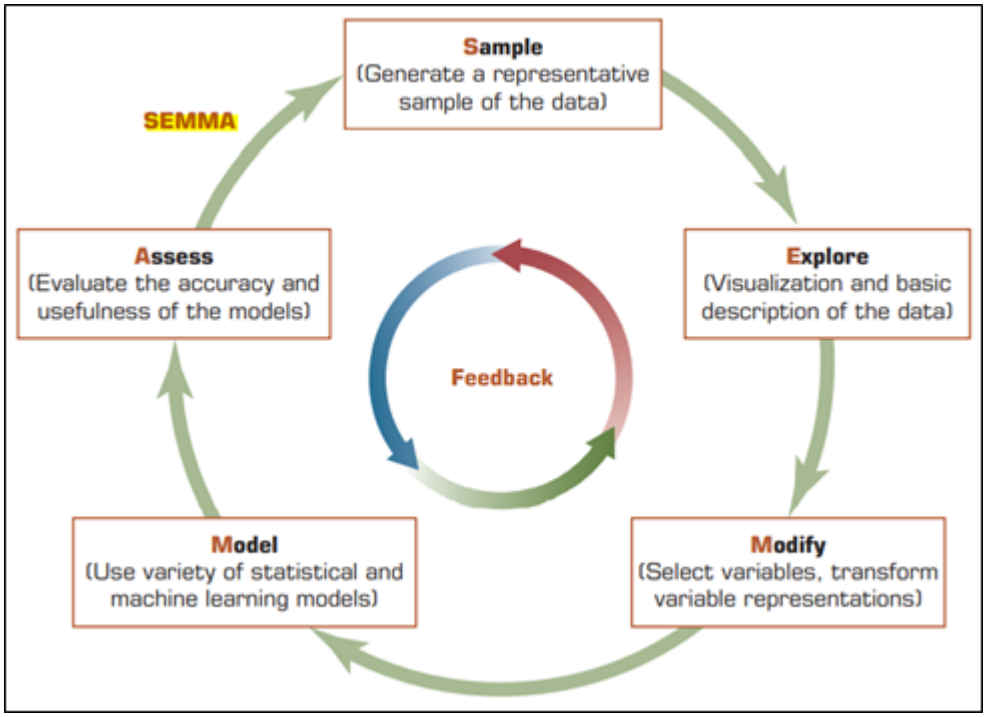


Table 1: CRISP-DM vs. SEMMA Comparison

SEMMA Step	Equivalent CRISP-DM	Focus
Sample	Data Understanding	Identifying the right dataset.
Explore	Data Understanding	Looking for outliers and abnormalities.
Modify	Data Preparation	Creating new variables, normalization.
Model	Modeling	The actual mining step.
Assess	Evaluation	Comparing model accuracy.

3: Data Types and Preprocessing for Modeling

2.1 Classification of Variables

A. Categorical (Qualitative) Variables

- **Nominal:** Categories with no natural order. (e.g., *State* or *Area Code* in the Churn dataset).
- **Ordinal:** Categories with a meaningful rank. (e.g., *Customer Service Calls* if treated as Low/Medium/High).

B. Numerical (Quantitative) Variables

- **Continuous:** Can take any value in a range (e.g., *Income* in the Adult dataset).
- **Discrete:** Countable values (e.g., *Number of VMail Messages*).

2.2 Handling Abnormalities and Missing Data

Chapter 2 emphasizes that "Dirty Data" is the norm.

- **Missing Values:** We must decide whether to delete records (dropping rows) or impute values (replacing missing values with the mean/median).
- **Data Abnormalities:** In the *Churn* dataset, if *Area Code* and *State* don't align logically, it suggests a data entry error that must be resolved before modeling.

2.3 Relationship between Variables: The Target Variable

In the *Adult* dataset, the **Target Variable** is *Income* ($> \$50k$ or $\leq \$50k$).

- We use **Bar Charts with Overlays** to see how categorical variables (like *Education*) affect the target.
- **Significance:** Variables that show a clear difference in the overlay (e.g., more high-income earners in the "Doctorate" category) are likely to be "significant" in a classification model.

4: Descriptive Statistics

Descriptive statistics provide a summary of the data, helping us understand its "shape" before we apply complex mining models. These measures describe where the data is centered and how much it varies.

Measures of Central Tendency

These indices describe the **center** of a distribution:

- **Mean ($\bar{x}$):** The arithmetic average. It is highly sensitive to outliers.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- **Median:** The middle value in a sorted list. It is **robust**, meaning it is not easily influenced by extreme values. In skewed distributions, such as income, the median is a more "typical" representation than the mean.
- **Mode:** The most frequent value. This is the only measure of center usable for **Nominal** categorical variables (e.g., the most common state in a customer dataset).

Measures of Variability (Spread)

Knowing the center is insufficient; we must understand the "dispersion" or "spread" of the data points.

- **Range:** The difference between the maximum and minimum values ($X_{max} - X_{min}$). While simple, it is entirely dependent on extreme outliers.
- **Standard Deviation (s):** Quantifies the average distance of every data point from the mean. A higher s indicates higher variability and "noisier" data.

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

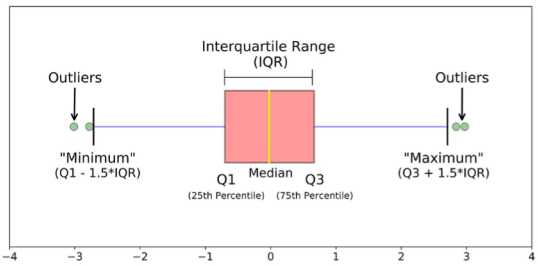
- **Interquartile Range (IQR):** The range of the middle 50

5: Outliers and Outlier Detection

An **outlier** is an observation that lies an abnormal distance from other values in a random sample from a population. Outliers can stem from data entry errors (e.g., a weight of 500 kg for a human) or represent true but rare anomalies (e.g., high-value fraud).

Graphical Method: The Box Plot

The Box Plot (or Whisker Plot) is the primary visual tool for outlier detection using the **1.5 × IQR Rule**. A value is mathematically flagged as an outlier if it falls outside the "fences":



1. **Upper Outliers:** Values $> Q3 + (1.5 \times IQR)$
2. **Lower Outliers:** Values ¡ $Q1 - (1.5 \times IQR)$

Numerical Method: The Z-Score

The **Z-Score** (standard score) indicates how many standard deviations a value is from the mean.

$$Z = \frac{x - \bar{x}}{s}$$

- A Z-score of 0 indicates the value is exactly the average.
- In a normal distribution, 99.7

6: Data Normalization Techniques

Most data mining algorithms (e.g., k-Nearest Neighbor or Neural Networks) rely on "distance" calculations between records. If variables have different scales (e.g., **Age** 0–100 vs. **Annual Income** 0–200,000), the variable with the larger range will dominate the model. **Normalization** ensures all variables contribute equally.

Min-Max Normalization

This technique scales the data into a fixed range, typically **[0, 1]**.

$$X_{norm} = \frac{X - min(X)}{max(X) - min(X)}$$

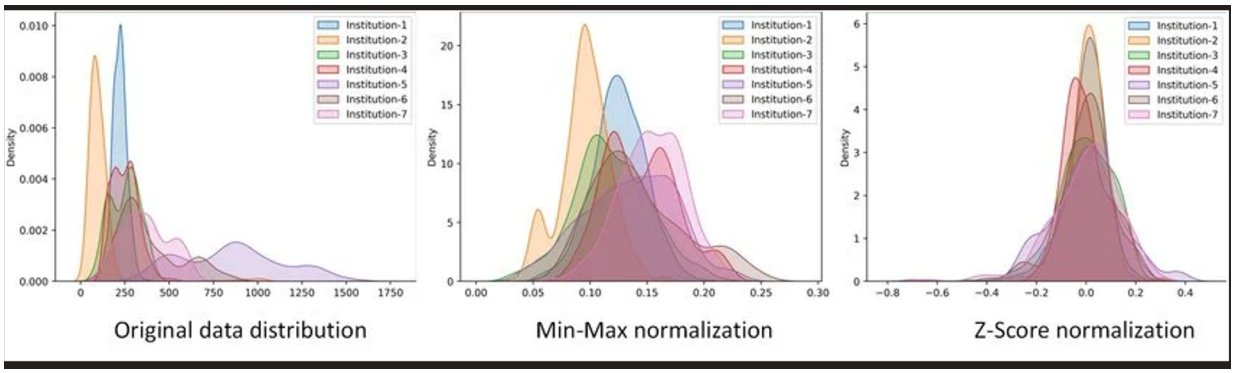
- **Advantage:** It preserves the exact relationships between the original data values.
- **Disadvantage:** It is highly sensitive to outliers; one extreme value can "squash" all other data points into a tiny range near zero.

Z-Score Standardization

This transforms the data so that it has a **Mean of 0** and a **Standard Deviation of 1**.

$$Z = \frac{x - \mu}{\sigma}$$

- **Advantage:** It is less affected by outliers than Min-Max normalization.
- **Usage:** Preferred when the data follows a roughly normal distribution or when the absolute bounds (min/max) are unknown.



Decimal Scaling and Log Transformations

- **Decimal Scaling:** Moves the decimal point of values so the maximum absolute value is less than 1. For example, if the max value is 999, we divide all values by 10^3.
- **Log Transformation:** Used to achieve **Normality**. If data is highly "right-skewed" (like population or wealth), taking the $log(x)$ pulls the outliers closer to the center, making the distribution more symmetric and suitable for statistical modeling.